FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2021

PAPER –2

TEST DATE: 26-09-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

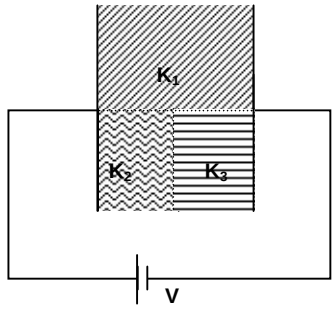
SECTION – A

1. A, C, D

Sol. $C_1 = \dfrac{K_1\varepsilon_0 A/2}{d} = \dfrac{\varepsilon_0 A}{d}$

$C_2 = \dfrac{K_2\varepsilon_0 A/2}{d/2} = \dfrac{3\varepsilon_0 A}{d}$

$C_3 = \dfrac{K_3\varepsilon_0 A/2}{d/2} = \dfrac{6\varepsilon_0 A}{d}$



Equivalent of C_2 and C_3

$C_{23} = \dfrac{C_2 \times C_3}{C_2 + C_3} = \dfrac{2\varepsilon_0 A}{d}$

Ratio of charges on the upper and lower half of one of the plates

$\dfrac{Q_1}{Q_2} = \dfrac{C_1}{C_{23}} = \dfrac{1}{2}$ (As C_1 and C_{23} are in parallel)

$E_1 = \dfrac{Q_1}{k_1 \dfrac{A}{2}\varepsilon_0} = \dfrac{Q_1}{A\varepsilon_0} = \dfrac{Q_2}{2A\varepsilon_0}$

$E_2 = \dfrac{Q_2}{k_2 \dfrac{A}{2}\varepsilon_0} = \dfrac{2Q_2}{3A\varepsilon_0}$, $E_3 = \dfrac{Q_2}{k_3 \dfrac{A}{2}\varepsilon_0} = \dfrac{Q_2}{3A\varepsilon_0}$

$E_1 : E_2 : E_3 = 1 : \dfrac{4}{3} : \dfrac{2}{3} = 3 : 4 : 2$

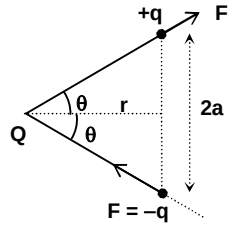
Ratio of energy densities in dielectrics

$= \frac{1}{2}k_1\varepsilon_0E_1^2 : \frac{1}{2}k_2\varepsilon_0E_2^2 : \frac{1}{2}k_3\varepsilon_0E_3^2 = 3 : 8 : 4$

2. A, D

Sol. $F_{net} = 2F\sin\theta$

$= 2q\frac{kQ}{(r^2+a^2)}\frac{a}{\sqrt{r^2+a^2}} = \frac{kQp}{\left(r^2+a^2\right)^{3/2}} = \frac{kQp}{r^3}$ (since r>>a)



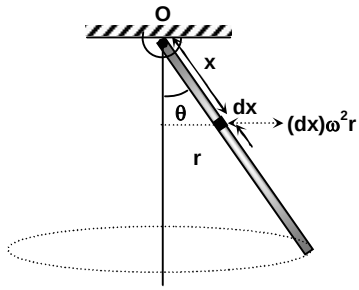
$\tau = 2qE\cos\theta a = 2q\frac{kQ}{(r^2+a^2)}\frac{r}{\sqrt{r^2+a^2}}a$

$= \frac{kQpr}{\left(r^2+a^2\right)^{3/2}} = \frac{kQp}{r^2}$ (since r>>a)

3. B, C

Sol. Considering a small element of length 'dx' of the rod

Torque of centrifugal force on the rod



$\tau = \int(dm)\omega^2rx\cos\theta = \int\left(\frac{m}{\ell}dx\right)\omega^2x^2\sin\theta\cos\theta$

$= \frac{m\omega^2\sin2\theta}{2\ell}\int_0^{\ell}x^2dx = \frac{m\omega^2\ell^2\sin2\theta}{6}$

Since, in the reference frame attached with the rod, the rod is in equilibrium

$\Rightarrow \quad \frac{m\omega^2\ell^2\sin2\theta}{6} = \frac{mg\ell\sin\theta}{2} \Rightarrow \omega = \sqrt{\frac{3g}{2\ell\cos\theta}}$

As centre of mass has no vertical acceleration, vertical component of reaction force = mg

Let F_x be the horizontal component of reaction force at O

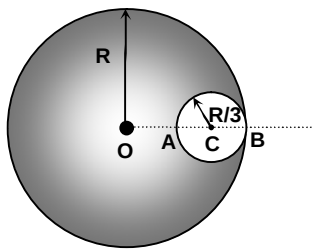
$F_x = ma_{cm} = m\omega^2\frac{\ell}{2}\sin\theta$

Angular momentum of the rod about axis of rotation

$L = \int(dm)\omega r^2 = \int_0^{\ell}\left(\frac{m}{\ell}dx\right)\omega x^2\sin^2\theta = \frac{(m\omega\sin^2\theta)\ell^2}{3}$

4. A, D

Sol. If P is any point the cavity



$\vec{E}_P = \frac{\rho}{3\varepsilon_0}(\overrightarrow{OP}) - \frac{\rho}{3\varepsilon_0}(\overrightarrow{CP})$

$= \frac{\rho}{3\varepsilon_0}(\overrightarrow{OP} - \overrightarrow{CP}) = \frac{\rho}{3\varepsilon_0}(\overrightarrow{OC})$

So, electric field in the cavity is uniform and directed along $\overrightarrow{OC}$

$E_C = \frac{\rho}{3\varepsilon_0}\left|\overrightarrow{OC}\right| = \frac{\rho}{3\varepsilon_0}\left(R - \frac{R}{3}\right) = \frac{2\rho R}{9\varepsilon_0}$

$v_A - v_C = v_C - v_B = E\frac{R}{3}$

E_0 = electric field at O due to entire spherical charge-electric field due to the charge which is removed

$$= 0 - \frac{1}{4\pi\varepsilon_0}\frac{\rho \times \frac{4}{3}\pi\left(\frac{R}{3}\right)^3}{\left(\frac{2R}{3}\right)^2} = \frac{\rho R}{36\varepsilon_0}$$

5. A, C, D

Sol. Frequency of sound coming from the source received by the observer

$$f_1 = \frac{(330+38)}{(330+10)} \times 340 = 368 \text{ Hz}$$

Frequency of sound coming to the reflector

$$f_r = \frac{(330-15)}{(330-10)} \times 340 = \frac{315}{320} \times 340 \text{ Hz}$$

Frequency of echo heard by the observer

$$f_2 = \frac{(330+38)}{(330+15)} \times \frac{315}{320} \times 340 = 357 \text{ Hz}$$

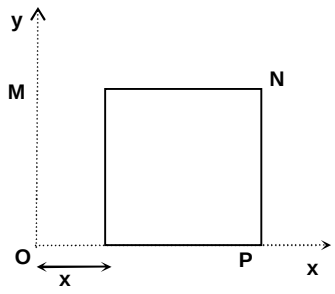
Apparent wavelength of echo heard by the observer

$$\lambda_{app} = \frac{V_{app}}{f_2} = 1.03 \text{ m}$$

6. A, D

Sol. If v be the velocity of the loop at any time t induced emf in the loop

$$\varepsilon = B_0\left(1+\frac{x+a}{a}\right)av - B_0\left(1+\frac{x}{a}\right)av$$ (Here x is the distance of the loop from the y-axis at time t)



$\Rightarrow \varepsilon = B_0av$

Induced current in the loop $I = \frac{\varepsilon}{R} = \frac{B_0av}{R}$

Magnetic force on the loop

$$F = IaB_0\left(1+\frac{x}{a}\right) - IaB_0\left(1+\frac{x+a}{a}\right)$$

$$= -IaB_0 = \frac{-B_0^2a^2v}{R}$$

$F = ma$

$$\Rightarrow \frac{-B_0^2a^2v}{R} = m\frac{dv}{dt}$$

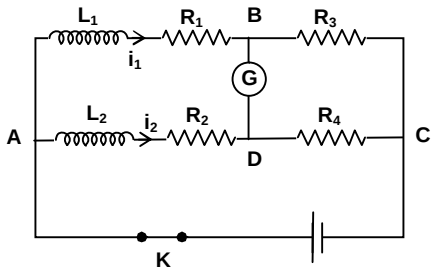
$$\Rightarrow \int_{v_0}^{v}\frac{dv}{v} = -\frac{B_0^2a^2}{mR}\int_0^t dt$$

$$\Rightarrow v = v_0e^{-\frac{B_0^2a^2t}{mR}}$$

SECTION – B

7. 2

Sol. Since there is no current through BD therefore points B and D are at same potential



$V_B = V_D$

Potential difference, $V_{AB} = V_{AD}$

$$L_1\frac{di_1}{dt} + i_1R_1 = L_2\frac{di_2}{dt} + i_2R_2 \quad \ldots(i)$$

Similarly,

$V_{BC} = V_{DC}$

$$i_1R_3 = i_2R_4 \quad \ldots(ii)$$

from equation (i) and (ii)

$$L_1\frac{R_4}{R_3}\frac{di_2}{dt} + i_2\frac{R_4R_1}{R_3} = L_2\frac{di_2}{dt} + i_2R_2$$

$$\left(L_1\frac{R_4}{R_3} - L_2\right)\frac{di_2}{dt} = i_2\left[R_2 - \frac{R_4R_1}{R_3}\right] \quad \ldots(iii)$$

At $t = 0$, $i_2 = 0$

$$L_1\frac{R_4}{R_3} - L_2 = 0 \text{ as } \frac{di_2}{dt} \neq 0$$

So, $\frac{L_1}{L_2} = \frac{R_3}{R_4}$

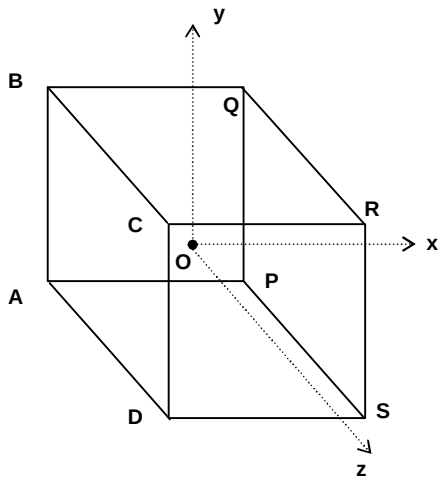
8. 4

Sol. $\phi_{net} = \phi_{ABCD} + \phi_{PQRS} + \phi_{BCRQ} + \phi_{ADSP}$

$= (-4\times0.5)\times1^2\cos180° + (4\times0.5)\times1^2\times\cos0°$

$$-3\left(\frac{1}{4}+2\right)\times1^2\times\cos0° - 3\left(\frac{1}{4}+2\right)\times1^2\times\cos180° = 4 \text{ N-m}^2/\text{C}$$

$q_{net} = \varepsilon_0\phi_{net} = 4\varepsilon_0$



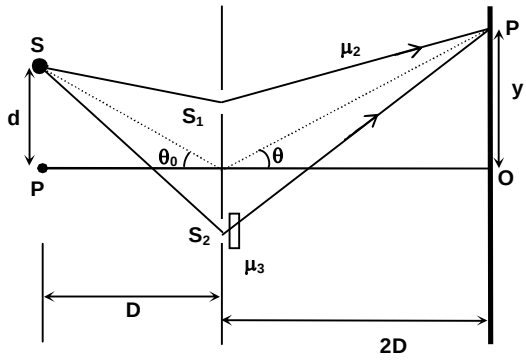
9. 7

Sol. Path difference at a point P on the screen

$$\Delta r = \mu_2 d\sin\theta_0 + \mu_2(S_2P - t) + \mu_3 t - \mu_2 S_1P$$

$$= \mu_2 d\tan\theta_0 + \mu_2(S_2P - S_1P) + (\mu_3 - \mu_2)t$$

$$= \mu_2 d\frac{d}{D} + \mu_2 d\frac{y}{2D} + (\mu_3 - \mu_2)t$$



Central bright fringe will be formed at P

If $\Delta r = 0$

$$\Rightarrow \mu_2\frac{d^2}{D} + \mu_2 d\frac{y}{2D} + (\mu_3 - \mu_2)t = 0$$

$$\Rightarrow \mu_2 d\frac{y}{2D} = -\mu_2\frac{d^2}{D} - (\mu_3 - \mu_2)t$$

$$\frac{1.2\times2\times10^{-3}}{2}y = \frac{-1.2\times4\times10^{-6}}{1} - 0.6\times6\times10^{-6}$$

$$\Rightarrow 1.2\times10^{-3}y = -8.4\times10^{-6}$$

$$\Rightarrow y = -\frac{8.4}{1.2}\times10^{-3}\text{m} = -7\text{ mm}$$

So, distance of the central bright fringe from O is 7 mm

10. 5

Sol. Initial potential difference between shells = $\frac{kq}{R} - \frac{kq}{2R} = \frac{kq}{2R}$ = 10 volt

Let q′ be the change on the outer shell after switch S_1 is closed

Then $k\left(\frac{q}{2R} + \frac{q'}{2R}\right) = 0 \Rightarrow q' = -q$

If q_1 be the charge on inner shell after switch S_1 is opened and S_2 is closed

$$k\left(\frac{q_1}{R} + \frac{-q}{2R}\right) = 0 \Rightarrow q_1 = +\frac{q}{2}$$

After S_1 and S_2 are closed alternatively n time charge on the inner shell

$$q_n = \frac{q}{2^n}$$

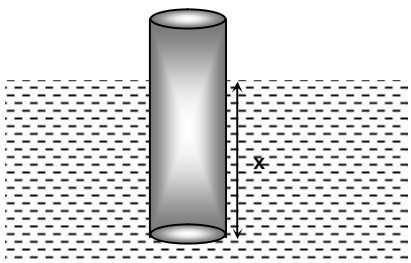
So, potential difference between the shells

$$v = kq_n\left(\frac{1}{R} - \frac{1}{2R}\right) = \frac{k}{2R}\frac{q}{2^n} = \frac{kq}{2^{n+1}R}$$

When n = 3, $V = \frac{1}{8}\left(\frac{kq}{2R}\right) = \frac{5}{4}$ volt

11. 3

Sol. Let at a certain moment height of immersed part of the cylinder be x



$$W_B + W_g = K_f - K_i$$

$$\int_L^0(-A\rho gx\,dx) - mgL = \frac{1}{2}mv^2 - 0$$ (ρ = density of the water)

$$\Rightarrow A\rho g\frac{L^2}{2} - mgL = \frac{1}{2}mv^2$$

$\Rightarrow v = \sqrt{\left(\frac{A\rho L}{m} - 2\right)gL} = \sqrt{\left(\frac{A\rho L}{A\rho_0 L} - 2\right)gL}$ (ρ_0 = density of the cylinder)

$= \sqrt{\left(\frac{1000}{300} - 2\right) \times 10 \times 67.5 \times 10^{-2}} = 3$ m/s

12. 9

Sol. $r = 60° - \theta_C$

$2\{i - (60° - \theta_C)\} + 180° - 2\theta_C = 108°$

$\Rightarrow i = 24°$

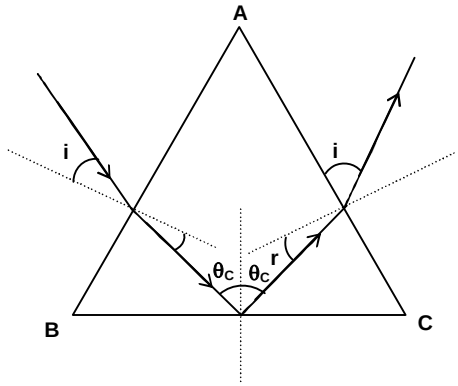
From snell's law

$1 \times \sin i = \mu \sin(60° - \theta_C)$

$= \mu\left[\frac{\sqrt{3}}{2}\cos\theta_C - \frac{1}{2}\sin\theta_C\right] = \mu\left[\frac{\sqrt{3}}{2}\sqrt{1 - \frac{1}{\mu^2}} - \frac{1}{2} \times \frac{1}{\mu}\right]$

$0.4 = \frac{\sqrt{3}}{2}\sqrt{\mu^2 - 1} - \frac{1}{2}$

$\sqrt{\mu^2 - 1} = \frac{9}{5\sqrt{3}}$



SECTION – C

13. 00000.50

Sol. Since the cylinder moves very slowly, it remains in equilibrium.

$\Rightarrow F + f\frac{3}{5} = N \times \frac{4}{5}$

Balancing torque about centre

$Fr = fr \Rightarrow f = F$

Further $F\left(r + \frac{3r}{5}\right) = mg\frac{4r}{5}$

$F = \frac{mg}{2}$

$\Rightarrow f = \frac{mg}{2}$

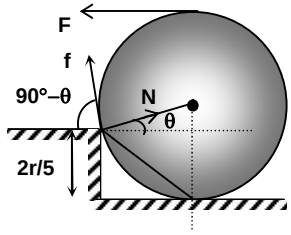
$N = \frac{5}{4}\left(F + \frac{3f}{5}\right) = \frac{5}{4}\left(\frac{mg}{2} + \frac{3mg}{10}\right) = mg$

The cylinder will not slip

If $f \le \mu N$

$\Rightarrow \frac{mg}{2} \le \mu mg \Rightarrow \mu \ge \frac{1}{2}$

$\Rightarrow \mu_{min} = \frac{1}{2}$



14. 00018.45

Sol. Work done by the gas = work done against the force of atmospheric pressure + work done against the weight of the mercury column

$= P_0 A \times \frac{5\ell}{2} + (U_f - U_i)$

$$= \frac{5}{2}P_0A\ell + 2A\ell\rho_m g\left(\ell - \frac{\ell}{8}\right)$$

$$= \frac{5}{2}P_0A\ell + \frac{7}{4}A\ell^2\rho_m g = 18.45 \text{ J}$$

15. 00000.25

Sol. If v_0 is the speed of the particle when it reaches the centre of the earth, from conservation of mechanical energy

$$\frac{1}{2}m(gR) - \frac{GMm}{R} = \frac{1}{2}mv_0^2 - \frac{3GMm}{2R}$$

$$\Rightarrow \frac{1}{2}mgR - mgR = \frac{1}{2}mv_0^2 - \frac{3}{2}mgR \quad (\text{since } g = \frac{GM}{R^2})$$

$$\Rightarrow v_0 = \sqrt{2gR}$$

$$\Rightarrow v = v_0\cos\omega t$$

$$\Rightarrow \sqrt{gR} = \sqrt{2gR}\cos\left(\sqrt{\frac{g}{R}}t\right)$$

$$\Rightarrow \sqrt{\frac{g}{R}}t = \frac{\pi}{4}$$

$$\Rightarrow t = \frac{\pi}{4}\sqrt{\frac{R}{g}}$$

16. 00000.75

Sol. $$F\cos\theta - mg - \mu N = m\frac{dv}{dt}$$

$$\Rightarrow F\cos\theta - mg - \mu F\sin\theta = m\frac{dv}{dt}$$

$$\Rightarrow m\int_{\theta}^{0} dv = \int_{0}^{\pi/2k} (F\cos kt - mg - \mu F\sin kt)dt$$

When $\theta = \frac{\pi}{2}$, $kt = \frac{\pi}{2}$ and $t = \frac{\pi}{2k}$

$$\Rightarrow F = \frac{\pi mg}{2(1-\mu)} = \frac{\pi mg}{2\left(1-\frac{1}{3}\right)} = \frac{3\pi mg}{4}$$

17. 00000.26

Sol. Let A_1 and A_2 be the initial activities of P^{32} and P^{33}

$$\frac{A_1}{A_2} = \frac{\lambda_1 N_1}{\lambda_2 N_2} = \frac{(t_{1/2})_2}{(t_{1/2})_1}\frac{N_1}{N_2} = \frac{25}{15}\times\frac{3}{2} = \frac{5}{2}$$

$$A_1 + A_2 = A_0 = 4.48 \text{ mCi}$$

$$\Rightarrow A_1 = 3.2 \text{ mCi and } A_2 = 1.28 \text{ mCi}$$

Activity of P^{32} after 75 days

$$A_1' = \frac{A_1}{2^5} = \frac{A_1}{32} = \frac{3.2}{32} = 0.10 \text{ mCi}$$

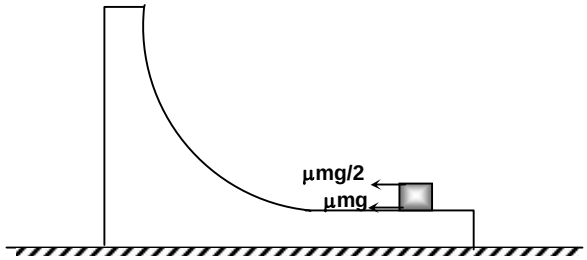
Activity of P^{33} after 75 days

$$A_2' = \frac{A_2}{2^3} = \frac{A_2}{8} = \frac{1.28}{8} = 0.16 \text{ mCi}$$

Total activity = $A_1' + A_2' = 0.26$ mCi

18. 00001.44

Sol. Let the velocity of block relative to ground when it comes to horizontal surface at B is v_1 and that of wedge is v_2



Using conservation of momentum

$mv_1 - 2mv_2 = 0$

$v_2 = \frac{v_1}{2}$

Using conservation of energy

$\frac{1}{2}mv_1^2 + \frac{1}{2}2m\left(\frac{v_1}{2}\right)^2 = mgr$

$v_1 = \sqrt{\frac{4gr}{3}}$, $v_2 = \frac{v_1}{2} = \sqrt{\frac{gr}{3}}$

Taking wedge as the reference frame, F.B.D. of block when it is on horizontal surface of the wedge is as shown.

$-\mu mg - \mu\frac{mg}{2} = ma$

$\Rightarrow a = \frac{-3\mu g}{2}$

$v^2 = u^2 + 2as$

$\Rightarrow 0 = \left(\frac{3v_1}{2}\right)^2 - 2 \times \frac{3\mu g}{2} \times \ell$

$\ell = \frac{3v_1^2}{4\mu g} = 2r = 1.44$ m

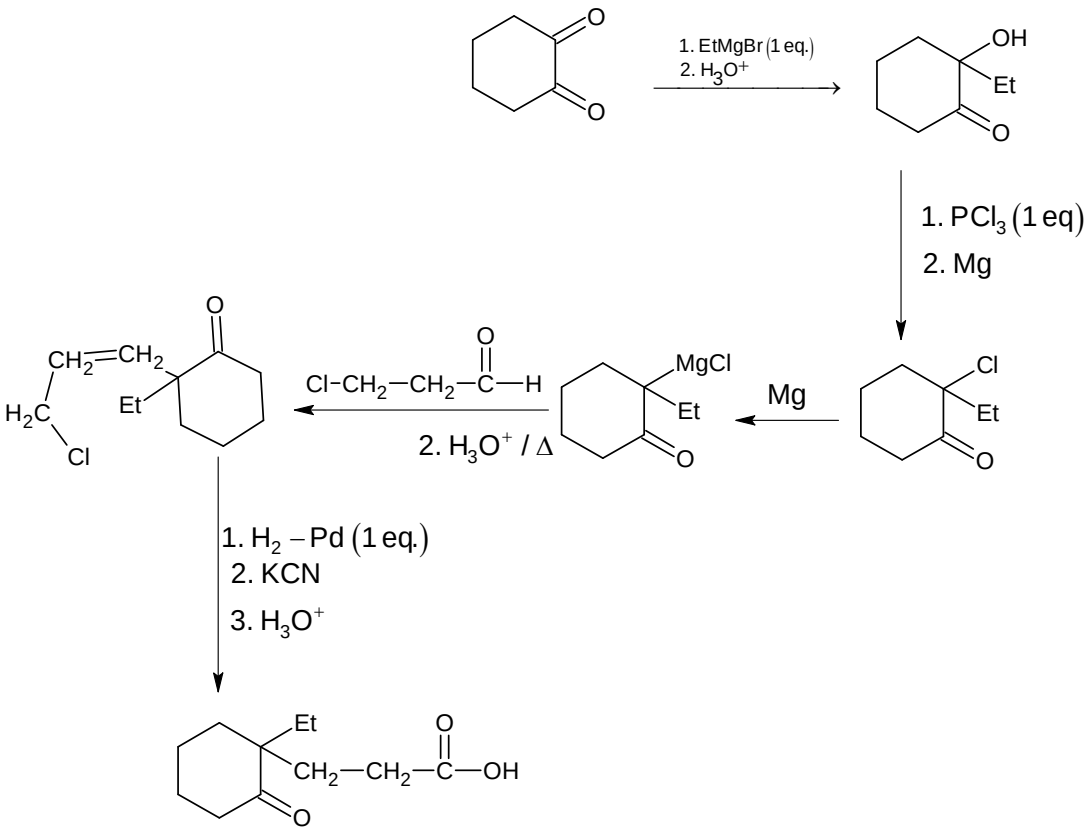
Chemistry

PART – II

SECTION – A

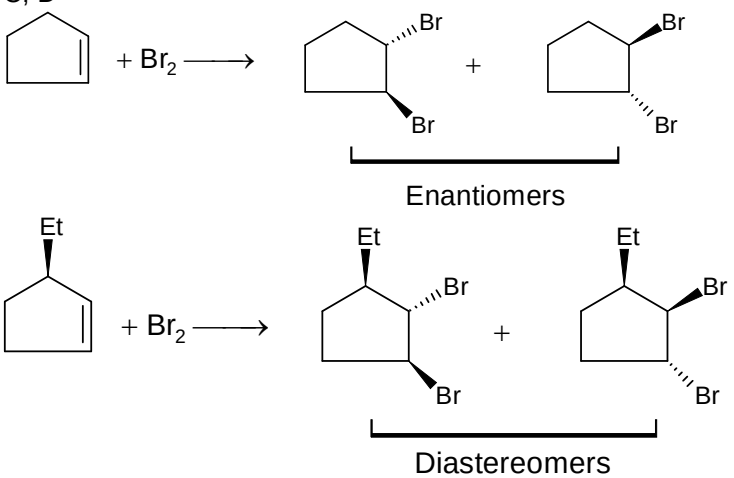
19. B, D

Sol.



20. C, D

Sol.



21. A, B, C

Sol. E_{cell}, E^{o}_{cell} and Normality are intensive properties.

22. A, C

Sol. $\left[Co(H_2O)_6\right]^{3+}$ and $\left[Fe(NH_3)_6\right]^{3+}$ are low spin complex. Where as $\left[Co(H_2O)_6\right]^{2+}$ and $\left[Co(H_2O)_3F_3\right]$ are high spin complex.

23. B, C, D

24. A, B, C, D

Sol. $XeF_6 + SiO_2 \longrightarrow XeOF_4 + SiF_4$

$XeOF_4 + H_2 \xrightarrow{300^oC} Xe + H_2O + HF$

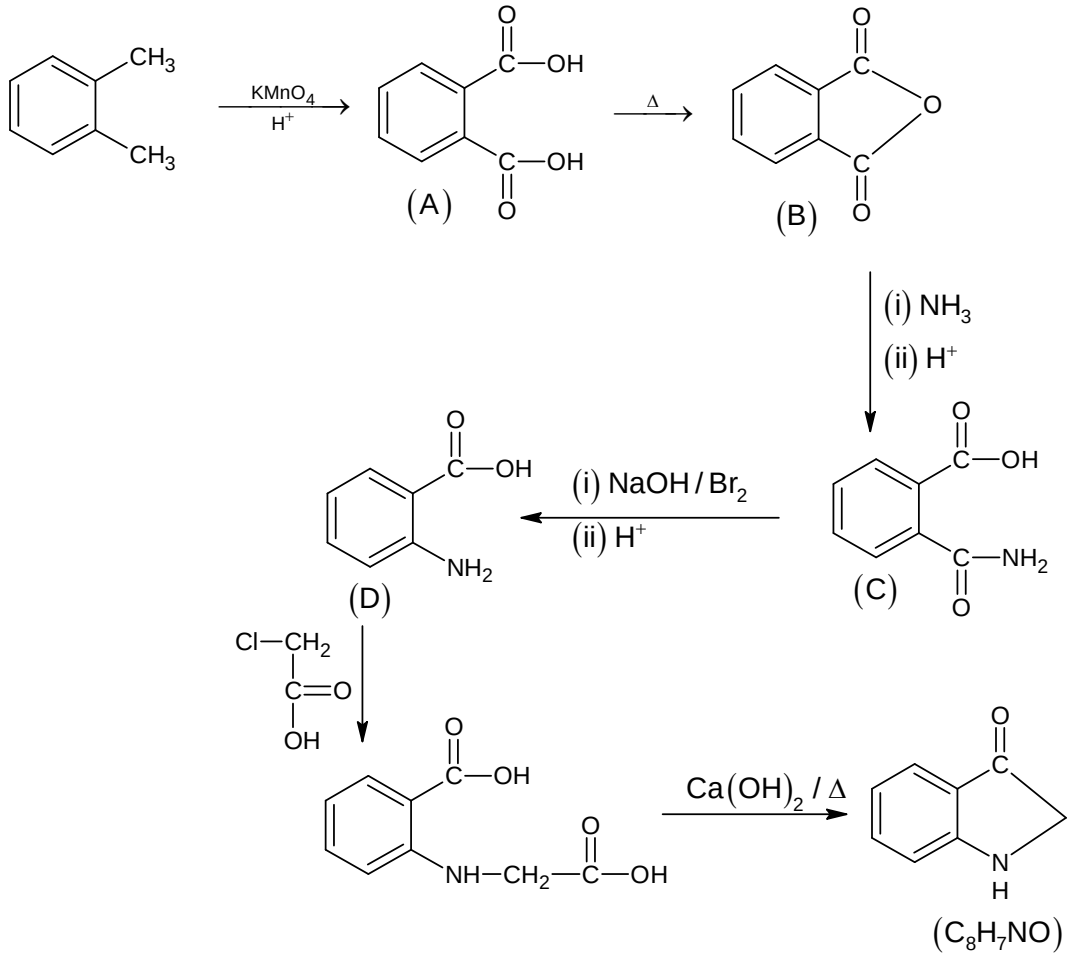
$XeO_2F_2 + H_2 \xrightarrow{300^oC} Xe + H_2O + HF$

XeO_3 disproportionate in strongly basic medium.

SECTION – B

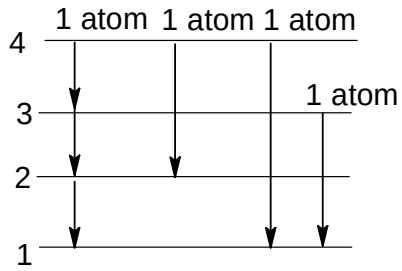
25. 6

Sol.



26. 4

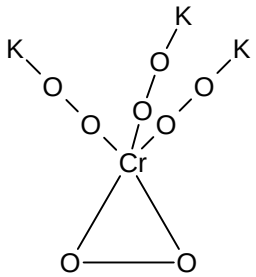
Sol. Number of spectral lines $= \frac{n(n-1)}{2} = \frac{4\times 3}{2} = 6$



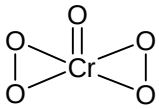
Total minimum atom required to get 6 different spectral line is 4.

27. 6

Sol.



Total peroxide linkage = 4



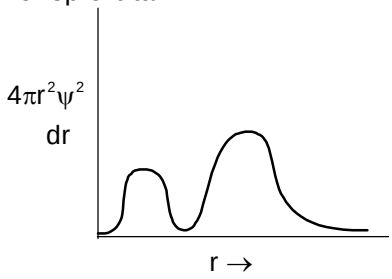
Total peroxide linkage = 2

$\therefore$ Sum of peroxide linkage = 2 + 4 = 6.

28. 4

Sol. All are correct.

For 3p-orbital.



29. 4

Sol. NiS, CoS, PbS and CuS are black colour sulphide.

30. 3

Sol. H_3PO_4, H_3BO_3 and H_3PO_3 these are ortho oxo acid.

SECTION – C

31. 00023.80

Sol. n = 1, f = 5

$$C_{V,m} = f\frac{R}{2} = 5\frac{R}{2}$$

$$dE = dq + dw$$

$dE + PdV = dq \Rightarrow dE + PdV = dq$

$\frac{dE}{dT} + \frac{PdV}{dT} = \frac{dq}{dT}$

$C_{v,m} + \frac{PdV}{dT} = \frac{dq}{dT} \Rightarrow \frac{dq}{dT} = \frac{5R}{2} + \frac{R}{3}$

$= \frac{15R + 2R}{6}$

$= \frac{17R}{6}$

Now, $\frac{P}{V^2} = 1$

$\Rightarrow P = V^2$

$PV = RT \Rightarrow V^3 = RT$

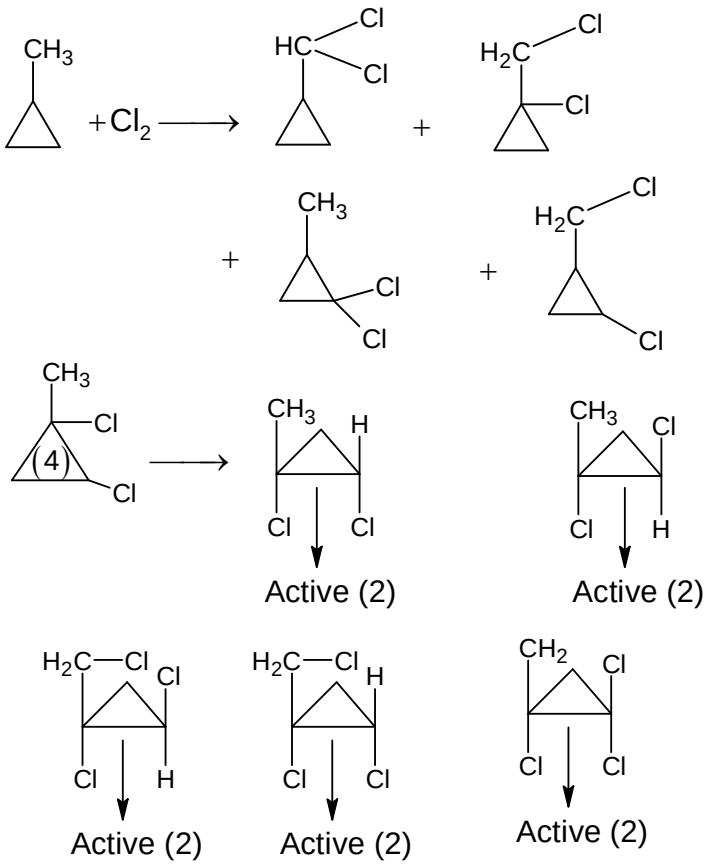
$\Rightarrow 3V^2 \frac{dV}{dT} = R$

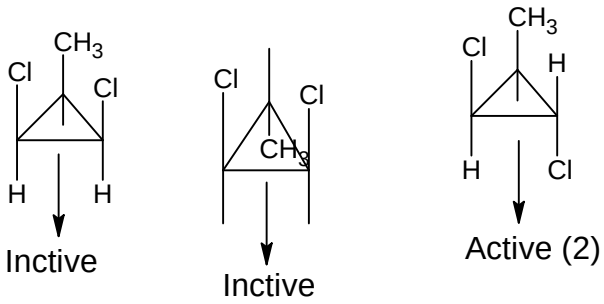
$\Rightarrow \frac{3PdV}{dT} = R \Rightarrow \frac{PdV}{dT} \Rightarrow \frac{PdV}{dT} = \frac{R}{3}$

$\therefore \frac{dq}{dT} = \frac{17R}{6}$

32. 00001.20

Sol.





$\therefore$ Value of x is 12.

$\therefore \dfrac{x}{10} = 1.2$.

33. 00000.20

Sol. $H_2S \rightleftharpoons 2H^+ + S^{2-}$

$$k_{a_1} \times k_{a_2} = \frac{\left[H^+\right]^2\left[S^{2-}\right]}{\left[H_2S\right]}$$

$$10^{-7} \times 1.6 \times 10^{-13} = \frac{\left[H^+\right]^2\left[S^{2-}\right]}{0.1}$$

$$10^{-7} \times 1.6 \times 10^{-13} \times 0.1 = \left[H^+\right]^2\left[S^{2-}\right] \quad \ldots (1)$$

Now, $MS \rightleftharpoons M^{2+} + S^{2-}$

$$K_{sp} = \left[M^{2+}\right]\left[S^{2-}\right]$$

$$10^{-21} \times 4 = 10^{-2} \times \left[S^{2-}\right]$$

$$4 \times 10^{-19} = \left[S^{2-}\right] \quad \ldots (2)$$

Now, $10^{-7} \times 1.6 \times 10^{-12} \times 0.1 = \left[H^+\right]^2\left[4 \times 10^{-19}\right]$

$$\therefore \left[H^+\right] = 0.2$$

$$= \frac{2}{10} = 0.2$$

34. 00000.05

Sol. $A_t = a_o \times e^{-[k_1+k_2+k_3+k_4]t} = 0$

$x + 2y + 3z + 4P = 0.1 \quad \ldots (1)$

$$\frac{x}{y} = \frac{1}{2}, \frac{y}{z} = \frac{2}{3}, \frac{z}{P} = \frac{3}{4}$$

$$x + 2[2x] + + \frac{6x}{2} + \frac{4(3x)}{3} = 0.1$$

$x + 4x + 9x + 6x = 0.1$

$$P = \frac{4 \times 3x}{3}$$

$= 16x$

$30x = 0.1$

$x = \frac{0.1}{300} = \frac{1}{300}$

$E_\infty = 4P = 16x = 16 \times \frac{1}{300} = \frac{16}{300}$

= 0.053 M

35. 00160.00

Sol. Molar mass of ethylene glycol = 62 g/mol

$\Delta T_f = K_f \times m$

$5 = 1.86 \times \frac{\frac{40}{62}}{W} \times 1000$

$5 = 1.86 \times \frac{40}{62} \times \frac{1000}{W}$

$W_{H_2O} = 1.86 \times \frac{40}{62 \times 5} \times 1000$

$W_{H_2O} = 240$ g

$\therefore$ The separated out = 400 – 240 = 160 g

36. 00000.01

Sol. $2H^+ + S^{2-}$

$H_2S \rightleftharpoons 2H^+ + S^{2-}$

$\frac{\left[H^+\right]^2\left[S^{2-}\right]}{\left[H_2S\right]} = K_{a_1} \times K_{a_2}$

$\frac{\left[10^{-4}\right]^2 \times \left[S^{2-}\right]}{0.01} = 10^{-7} \times 10^{-13} = 10^{-20}$

$\left[S^{2-}\right] = 10^{-14}$ M

Now, $Ag_2s \rightleftharpoons 2Ag^+ + S^{2-}$

$K_{sp} = \left[Ag^+\right]^2\left[S^{2-}\right]$

$4 \times 10^{-44} = \left[Ag^+\right]^2\left[10^{-14}\right]$

$\left[Ag^+\right] = 2 \times 10^{-15}$

Now, $Ag^+ + e^- \longrightarrow Ag$

$E_{Ag^+/Ag} = 0.9 - \frac{0.059}{1} \times \log\frac{1}{2 \times 10^{-15}}$

= 0.018

Mathematics

PART – III

SECTION – A

37. A, C, D

Sol. Q.M. $\geq$ A.M. $\geq$ G.M. $\geq$ H.M.

38. C, D

Sol. $(z+\bar{z}+2)^2+(z\cdot\bar{z}-6)^2=0$

$\Rightarrow z_1$ and z_2 will be $-1\pm i\sqrt{5}$

39. B, D

Sol. $f(i, j, k) = 0\ \forall\ i, j, k \in N$

40. C, D

Sol. Volume of MPSA = $\frac{1}{3}$ unit^3 and area of ΔBCD = 4 area of ΔRQC

41. A, B, C, D

Sol. $P=\frac{1}{8}$, $Q=\frac{1}{18}$, $R=\frac{\sin^2 1}{2}$

42. C, D

Sol. $A_n=\ell n\left(\frac{n^n}{n!}\right)$

SECTION – B

43. 3

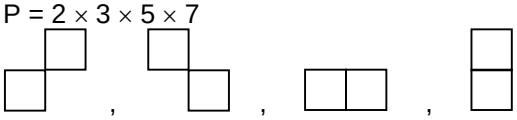
Sol. Period of recurrence is 5

44. 5

Sol. The domain of f(x) is $\left[-\frac{1}{2}\right]\left(\frac{45}{8}\right)-\{0\}$

45. 8

Sol. $P = 2\times3\times5\times7$



Possibility

46. 4

Sol. $(f(-1))^2 + f(1) \leq -2 \Rightarrow f(1) \leq -2$

$(f(1))^2 + f(1) \leq 2 \Rightarrow (f(1) + 2)(f(1) - 1) \leq 0$

$f(1) \in [-2, 1] \Rightarrow f(1) = -2, f(-1) = 0$

47. 0

Sol. $x + y + z = 0 \Rightarrow \sum x^3 = 3xyz$

$x^2 - y^2 = z = -(x + y) \Rightarrow (x + y)(x - y + 1) = 0$

if $x + y = 0$ and $z = 0 \Rightarrow$ if $x - y + 1 = 0 \rightarrow xyz = 0$

48. 1

Sol. Let $\theta = 10° \Rightarrow \frac{4\sin^2\theta}{\cos 3\theta} = \frac{1}{\cos 3\theta} - \frac{1}{\cos\theta}$

$\Rightarrow p = 12, q = 192$

SECTION – C

49. 00057.00

Sol. $k \in [22, 78]$

50. 00521.00

Sol. $\left(\frac{\sum a_i}{3}\right)^3 \geq (a_1 + a_4)(a_2 + a_5)(a_3 + a_6)$

51. 00057.00

Sol. $\sum a_1 = 3,\ \sum b_1 = 3 \Rightarrow \sum a_1^2 - \sum a_1 a_2 = \frac{20}{3},\ \sum b_1^2 - \sum b_1 b_2 = 7$

$(a_1 - a_2)^2 + (a_2 - a_3)^2 + (a_3 - a_1)^2 = \frac{40}{3}$

$(b_1 - b_2)^2 + (b_2 - b_3)^2 + (b_3 - b_1)^2 = 14$

$50a^2 = \frac{40}{3} + 14 \Rightarrow 6a^2 = \frac{82}{25} = \frac{p}{q}$

52. 00019.00

Sol. Multiply first equation by y and second by x

$2xy - 1 = 3y \Rightarrow x = \frac{3y+1}{2y}$

$\Rightarrow 4y^4 - 3y^2 - 1 = 0 \Rightarrow y = \pm 1,\ x = 2$ and 1

53. 00000.50

Sol. $\frac{AD}{AC} = \frac{\sin C}{\sin\left(\frac{A}{2} + B\right)},\ \frac{AE}{AC} = \cos\frac{A}{2}$

$\frac{AD}{AE} = \frac{2\sin C}{\sin C + \sin B} = \frac{1}{2}$

54. 00100.00

Sol. After squaring two times $(16 - 8p)t^2 = (p - 4)^2$

$\Rightarrow t = \frac{|P-4|}{\sqrt{16-8P}}$ put this in equation, then $P \in \left[0, \frac{4}{3}\right]$ for a solution